Sixth-order global reality and general even-order windows for arithmetic theta transforms

Project Stardust

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Abstract

We give a human proof of eventual reality and simplicity of every zero of the cosine transform obtained by multiplying the arithmetic theta kernel by $\exp(tu^6)$. The conclusion holds on the entire complex plane, with one sufficiently late time valid for all frequencies. The proof constructs actual stationary points in a moving transition region and its far complement, proves a uniform oscillatory Gaussian estimate for the full kernel, and converts an analytic cosine-pair approximation into exact real and simple zeros. The sixth-order polynomial changes, physical-prefix bound, and coverage by a common real stationary center are explicit. For every fixed integer-even order we additionally prove real and simple zeros inside an expanding saddle window, with full actual weighted-moment estimates and the precise non-global quantifier boundary. The results are matched to a frozen Lean development. No general integer-order classification, optimal threshold or Riemann-hypothesis consequence is asserted; novelty remains unresolved.

1 Statement and the actual integral

For $u \geq 0$ let

$$K(u) = \sum_{n\geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \qquad F_t(z) = \int_0^\infty e^{tu^6} K(u) \cos(zu)\, du. \tag{1}$$

The sign of the tilt is positive. Here and below t is real and z is complex. A zero is simple when $F_t'(z) \neq 0$.

Theorem 1.1. *There is a finite T such that, for every $t \geq T$ and every $z \in \mathbb{C}$, $F_t(z) = 0$ implies*

$$\operatorname{Im} z = 0, \qquad F_t'(z) \neq 0.$$

The time T does not depend on z. In particular, a fixed-frequency asymptotic cannot by itself prove this statement. We use one escaping stationary center to cover all frequencies. The proof follows the actual-saddle organization of the quartic companion [1], but supplies the sixth-order derivatives and estimates needed for the extension. It does not import the quartic arithmetic certificate or its endpoint collision.

Each summand in (1) is positive on the half-line. Summation after splitting off half the exponential gives

$$0 < K(u) \leq C e^{9u - \frac{\pi}{2} e^{4u}}, \qquad u \geq 0. \tag{2}$$

Indeed $n^2 e^{4u} \geq (n^2 + e^{4u})/2$ and $\sum n^4 e^{-\pi n^2/2} < \infty$. Thus every $u^k e^{tu^6 + Bu} K(u)$ is integrable, for fixed k, t, B. On compact parameter sets these are integrable dominating functions for all spatial

derivatives. Consequently F_t is entire, even, conjugation compatible, and nonzero since $F_t(0) > 0$. All contour integrals below are ordinary absolutely convergent integrals when their parameters are fixed.

2 The theta strip and the sixth-order slope

The kernel has an even holomorphic continuation to

$$\mathcal{U} = \{w : |\operatorname{Im} w| < \pi/8\}.$$

Here is the precise kernel information we shall use. Write $w = a + ib$, $E = e^{4w}$, $L = e^{4a}$ and $V = \operatorname{Re} E$. For $a \geq 0$, $V \geq 1$, normal convergence of the series gives

$$K(w) = 2\pi^2 e^{9w-\pi E}(1 + \rho(w)), \qquad |\rho^{(j)}(w)| \leq C_j\{e^{-4a} + (1+L)^j e^{-3\pi V}\}, \quad 0 \leq j \leq 4. \tag{3}$$

For completeness, after dividing by the leading exponential the remaining factor is

$$1 - \frac{3}{2\pi}e^{-4w} + \sum_{n\geq 2}\left(n^4 - \frac{3n^2}{2\pi}e^{-4w}\right)e^{-\pi(n^2-1)e^{4w}}.$$

Each of its first four derivatives is bounded by a constant times $(1+L)^j$ times a polynomial in n and the same exponential. For $n \geq 2$, $n^2 - 1 \geq 3$; splitting off this minimum and summing the remaining Gaussian proves (3).

The global strip bound needed at the physical endpoint is

$$|K(s + ib)| \leq C\cos(4b)^{-3}e^{-s}, \qquad s \geq 0. \tag{4}$$

To check it, compare $\sum n^m e^{-\pi V n^2}$, $m = 2, 4$, with the Gaussian integral from zero, obtaining $C_m V^{-(m+1)/2}$. Substitution of $V = e^{4s}\cos(4b)$ in the two terms of the kernel gives respectively $Ce^{-s}\cos(4b)^{-5/2}$ and $Ce^{-s}\cos(4b)^{-3/2}$, both bounded as stated.

Evenness follows from the Gaussian theta identity. If $\theta(v) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 v}$, its modular identity is $\theta(v) = v^{-1/2}\theta(1/v)$, with the right-half-plane square root; see [4]. Put $G(w) = e^w(\theta(e^{4w})-1)/2$. Then $G(-w) = G(w) + \sinh w$, and direct differentiation gives $(D^2 - 1)G = 8K$. This proves evenness throughout $\mathcal{U}$.

For large a, under $V \geq c_0 t a^5$, $t \geq 1$, the kernel is nonzero and its full logarithmic derivative is

$$\sigma_t(w) = 6tw^5 + \frac{K'(w)}{K(w)} = 6tw^5 + 9 - 4\pi e^{4w} + \epsilon(w), \qquad |\epsilon^{(j)}(w)| \leq Ce^{-4a} \quad (0 \leq j \leq 3). \tag{5}$$

This follows by differentiating $\log(1+\rho)$ in (3). Indeed $L^4 e^{-cV} \leq e^{16a - cta^5}$ is eventually smaller than e^{-4a}. The same bound holds along $w + s$, $s \geq -1$, with Ce^{-4a} on the right: the exponential terms are bounded by $\exp(16a + 16s - cta^5 e^{4s})$, whose maximum on this half-line is at its left endpoint for large a. All constants below are independent of t and frequency, unless a fixed parameter is indicated by a subscript.

3 Uniform analysis at an actual saddle

Fix $c_0 = 10^{-3}$. Call (t, z, v) admissible when

$$t \geq 1, \quad v = a + ib \in \mathcal{U}, \quad a \text{ large}, \quad y = \operatorname{Im} z \leq 1, \quad \sigma_t(v) + iz = 0, \quad V \geq c_0 t a^5. \tag{6}$$

These conditions will be proved in the constructions below; no theta asymptotic is applied at an assumed model root. Set

$$Y = \max(0, -y), \quad Q = ta^5 + Y, \quad \beta = -\sigma_t'(v)/2, \quad W_z(w) = e^{tw^6 + izw} K(w).$$

Taking real parts in full stationarity gives

$$4\pi V = 6t(a^5 - 10a^3b^2 + 5ab^4) + 9 - y + \operatorname{Re}\epsilon(v). \tag{7}$$

Since $|b| < \pi/8$, this implies, uniformly,

$$cQ \le V \le CQ, \quad Q \le CL, \quad Q \ge a^5, \qquad cL \le |\beta| \le CL, \quad cQ \le \operatorname{Re}\beta \le CQ. \tag{8}$$

In fact

$$\sigma_t'(v) = 30tv^4 - 16\pi e^{4v} + \epsilon'(v), \qquad \sigma_t''(v) = 120tv^3 - 64\pi e^{4v} + \epsilon''(v).$$

The polynomial terms have size at most $Cta^4 = o(Q)$, whereas the exponential has modulus L and real part V. On any sufficiently small fixed horizontal interval the second derivative is bounded by CL. On the entire horizontal ray $s \ge -1$ one has

$$\operatorname{Re}\sigma_t'(v+s) \le -cQ. \tag{9}$$

To see this, use $(a+s)^4 \le Ca^4e^{2s}$ for $s \ge -1$ and large a. Its contribution Cta^4e^{2s} is absorbed by $16\pi V e^{4s}$; their ratio is at most C/a on this ray. The error from (5) is negligible.

Proposition 3.1. *There is $\eta(a) \to 0$ such that, uniformly at admissible saddles,*

$$I_z(b) := \int_0^\infty W_z(s+ib)\,ds = W_z(v)\frac{\sqrt{\pi}}{\sqrt{\beta}}(1+\delta), \qquad |\delta| \le \eta(a), \tag{10}$$

$$\int_0^\infty |W_z(s+ib)|\,ds \le C|W_z(v)|/\sqrt{Q}. \tag{11}$$

The square root in (10) *has positive real part.*

Proof. On $s \ge -1$ put $P(s) = \int_0^s (\sigma_t(v+r) + iz)\,dr$. The exact scalar differential equation gives $W_z(v+s) = W_z(v)e^{P(s)}$. Full stationarity and (9) imply

$$\operatorname{Re}P(s) \le -cQs^2, \qquad P(s) = -\beta s^2 + O(L|s|^3) \quad (|s| \le \rho),$$

where $\rho > 0$ is fixed and small. In particular this argument does not assume a fixed complex disk as wide as the horizontal contour when b is near the strip boundary.

Let $r = L^{-7/16}$. The integral on $[-r, r]$ differs from the Gaussian integral there by $O(Lr^4)$, since $Lr^3 \to 0$ and $|e^h - 1| \le e^{|h|}|h|$. For $\operatorname{Re}\beta > 0$,

$$\int_{\mathbb{R}} e^{-\beta s^2}\,ds = \sqrt{\pi}/\sqrt{\beta}, \qquad \left|\int_r^\infty e^{-\beta s^2}\,ds\right| + \left|\int_{-\infty}^{-r} e^{-\beta s^2}\,ds\right| \le \frac{2}{|\beta| r}.$$

The identity follows by differentiating in β and integrating $(se^{-\beta s^2})'$: the product of the integral with $\sqrt{\beta}$ is constant and equals $\sqrt{\pi}$ on the positive real axis. Integration by parts using $(e^{-\beta s^2})' = -2\beta s e^{-\beta s^2}$ proves the tail bound. These are oscillatory bounds in $|\beta|$, not bounds that incorrectly compare its real part to its modulus.

On $r \le |s| \le \rho$, Taylor's theorem gives $|P'(s)| \ge cL|s|$ and $|P''(s)| \le CL$. Integration of $(e^P)'/P'$ bounds these annuli by $C/(Lr)$. The local remainder divided by the main Gaussian is $O(L^{3/2}r^4) = O(L^{-1/4})$, and each normalized Gaussian or annular tail is $O(L^{-1/16})$. The remaining intervals $[-1,-\rho]$ and $[\rho,\infty)$ are bounded by Ce^{-cQ}, using the real-part estimate. After normalization their bounds still tend to zero, since $\sqrt{L} = e^{2a}$ and $Q \ge a^5$.

It remains to treat the physical prefix $0 \le s \le a-1$. Write $z = X + iy$. Since $\cos(4b) = V/L \ge cQ/L$, (4) and $\mathrm{Re}(s+ib)^6 \le s^6 + C$ give

$$|W_z(s+ib)| \le Ce^{12a}Q^{-3}\exp\{t(a-1)^6 + Ct + Y(a-1) - Xb\}. \tag{12}$$

On the other hand, (3) and (7) give at the center

$$|W_z(v)| \ge \exp\{ta^6 - \tfrac{3}{2}ta^5 - Cta^4 + 9a + Y(a-\tfrac{1}{4}) - Xb - Ca\}. \tag{13}$$

Indeed $\mathrm{Re}\, v^6 = a^6 - 15a^4b^2 + 15a^2b^4 - b^6$ and $\pi V = \frac{3}{2}t\,\mathrm{Re}\, v^5 + (9 - y + \mathrm{Re}\,\epsilon)/4$. The difference of the polynomial exponents is

$$t\{(a-1)^6 - a^6 + \tfrac{3}{2}a^5 + Ca^4 + C\} = -\tfrac{9}{2}ta^5 + O(ta^4) \le -cta^5.$$

The Y difference is $-3Y/4$. Thus the prefix integral, divided by $|W_z(v)|$, is at most Cae^{-cQ+Ca}. It vanishes even after multiplication by $\sqrt{L}$. This completes (10). Absolute integration of e^{-cQs^2} and the same prefix bound proves (11). One may, for example, bound η by a constant times $L^{-1/4} + L^{-1/16} + (1+a)e^{-ca^5+Ca}$. □

4 Contours, reflection, and exact local zeros

Let $J_z = \int_0^\infty W_z(s)\,ds$. For real h in the theta strip, a rectangle gives

$$J_z = I_z(h) + i\int_0^h W_z(iy)\,dy.$$

At its right edge the modulus tends to zero: the leading term $-\pi e^{4R}\cos(4h)$ dominates $tR^6 + O(R^5)$ and all frequency and kernel polynomial terms. For $y = \mathrm{Im}\, z$ this yields the exact identities

$$F_t(z) = \frac{I_z(h) + \overline{I_{\bar z}(k)}}{2} + \frac{i}{2}\int_k^h W_z(iq)\,dq, \tag{14}$$

$$F_t(z) = \tfrac{1}{2}\{I_z(h) + I_{-z}(-h)\}. \tag{15}$$

Evenness of K and of the sixth power cancels the vertical integrals in the second formula.

For an admissible saddle with $y \le 0$, the reflected integrands satisfy the exact modulus relation

$$|W_{-z}(s-ib)| = e^{2ys}|W_z(s+ib)|.$$

On $s \ge a-1$ use $e^{2ys} \le e^{2y(a-1)}$ and (11). On the prefix use $e^{2ys} \le 1$ and the prefix bound. With $A_z = W_z(v)\sqrt{\pi}/\sqrt{\beta}$,

$$|I_{-z}(-b)/A_z| \le C\sqrt{L/Q}\,e^{2y(a-1)} + o(1). \tag{16}$$

In particular no such zero has $y \le -1$ for large a: $\sqrt{L} = e^{2a}$ cancels $e^{-2(a-1)}$, leaving $C/\sqrt{Q} + o(1)$, whereas $I_z(b)/A_z \to 1$.

Lemma 4.1. *Let f, E, Φ be holomorphic on $|z - x| \le 1$, x real, with conjugation compatibility and E nonvanishing. Suppose, for a large and $D/a \to 0$,*

$$|\Phi'(z) - a| \le D, \qquad |f(z)/E(z) - \cos\Phi(z)| \le \varepsilon \cosh(\operatorname{Im}\Phi(z)), \qquad \varepsilon \to 0.$$

Then every zero with $|z - x| < 1/2$ is real and simple.

Proof. At a zero q, write $\Phi(q) = u + iv$. The identity $|\cos(u + iv)|^2 = \cos^2 u + \sinh^2 v$ gives $\tanh|v| \le \varepsilon$ and $|\cos u| \le 2\varepsilon$. On the disk $|z - q| \le \delta/a$, $\delta = 10^{-3}$, integration of the derivative bound gives $|\Phi(z) - u| \le 2\delta + 2\varepsilon$. Take ε sufficiently small. If $d = \sin u$, then $|d| \ge 1/2$ and $|\sin\Phi(z) - d|$ is bounded by a small absolute constant. Cauchy's estimate on this disk bounds the derivative of $f/E - \cos\Phi$ on its half-disk by $C\varepsilon a/\delta$. Hence there

$$|(f/E)'(z) + ad| \le a|d|/2.$$

Also vertical integration of Φ' from the real axis gives $|\operatorname{Im}\Phi(q)| \ge (a - D)|\operatorname{Im} q|$. Thus $|q - \bar{q}| \le C\varepsilon/a < \delta/(2a)$. If q were nonreal, q and $\bar{q}$ would be distinct zeros in the half-disk. Integration of the last derivative bound along their segment gives $|(f/E)(q) - (f/E)(\bar{q})| \ge a|d||q - \bar{q}|/2 > 0$, a contradiction. The same bound shows the derivative at the real zero is nonzero. Multiplication by E preserves simplicity. □

Proposition 4.2. *If a real x has an admissible saddle $v_0 = a + ib$, then, for large a, every zero z with $|z - x| < 1/2$ is real and simple.*

Proof. The theta boundary has distance at least cQ/L from v_0, because $\cos(4b) \ge cQ/L$. This is much larger than $1/L$. For $|z - x| \le 2$, solve $\sigma_t(v) + iz = 0$ by the contraction

$$v \longmapsto v - \frac{\sigma_t(v) + iz}{\sigma_t'(v_0)}$$

on a disk of radius C/L about v_0. Its derivative has modulus $O(1/L)$, by the CL bound on σ_t'', and its center displacement is $O(1/L)$. It gives an analytic branch with $v(z) = v_0 + O(1/L)$ and $v'(z) = -i/\sigma_t'(v(z)) = O(1/L)$. For $|z - x| \le 1$, the perturbed saddle remains admissible: first its margin is at least $(c_0/2)ta^5$, then (7) restores a fixed stronger margin for large a. The quantities a, L, Q remain comparable to their central values.

The nonzero analytic function $A(z) = W_z(v(z))\sqrt{\pi}/\sqrt{\beta(z)}$ has logarithmic derivative

$$\frac{A'(z)}{A(z)} = iv(z) - \frac{\beta'(z)}{2\beta(z)} = iv(z) + O(1/L). \tag{17}$$

Full stationarity cancels the derivative of v in the exponential weight. Also $|\beta'| = O(1)$ because $|\sigma_t''| = O(L)$ and $|v'| = O(1/L)$. Choose $P = \log(A/A(x))$ with $P(x) = 0$, put $P^*(z) = \overline{P(\bar{z})}$ and fix a real argument α of $A(x)$. Define

$$E(z) = |A(x)|e^{(P+P^*)/2}, \qquad \Phi(z) = \alpha + (P - P^*)/(2i).$$

Both have the required conjugation property and $\Phi' = a + O(1/L)$. Exactly,

$$\tfrac{1}{2}(A(z) + \overline{A(\bar{z})}) = E(z)\cos\Phi(z), \qquad \tfrac{1}{2}(|A(z)| + |A(\bar{z})|) = |E(z)|\cosh(\operatorname{Im}\Phi(z)).$$

Use (14) with heights $h = \operatorname{Im} v(z)$ and $k = \operatorname{Im} v(\bar{z})$. Proposition 3.1 controls its two horizontal integrals by these A terms. The heights differ by $O(|\operatorname{Im} z|/L)$, and throughout their intervening

interval $\cos(4q) \geq cQ/L$. Equations (4) and $\mathrm{Re}(iq)^6 = -q^6 \leq 0$ give a bound $\exp(Ca+t-Xh+C)$ for the residual vertical integral, where $X = x$ may replace $\mathrm{Re}\, z$ at bounded cost. At the central real saddle, stationarity implies $|x| \leq CL$, so $|X(h-k)| = O(1)$. By (13), meanwhile, $|A(z)| \geq \exp(ta^6 - Cta^5 - Ca - Xh)$. The ratio tends uniformly to zero since $t \geq 1$ and $a \to \infty$. Thus

$$|F_t(z)/E(z) - \cos\Phi(z)| \leq o(1)\cosh(\mathrm{Im}\,\Phi(z)), \qquad |z-x| \leq 1.$$

Lemma 4.1 applies with $D = C/L$. □

5 The moving transition region

The positive continuous function $e^{tu^6}K(u)$ tends to zero at infinity and attains a maximum on $[0,\infty)$. Every such maximizer $u = u_t$ escapes to infinity as $t \to \infty$. Indeed for a fixed R, the values on $[0,R]$ are at most $C_R e^{tR^6}$, whereas the value at $R+1$ is $K(R+1)e^{t(R+1)^6}$ and eventually larger. At a positive maximizer differentiation gives $\sigma_t(u) = 0$. With $B = 4\pi e^{4u}$,

$$B = 6tu^5 + 9 + \epsilon(u) \asymp tu^5. \tag{18}$$

This is the center used in both frequency regions.

Proposition 5.1. *Fix $M > 0$. For all sufficiently large t, every zero of F_t in $|z| \leq MB$ is real and simple.*

Proof. It suffices first to take $\mathrm{Im}\, z \leq 0$. The anchor $q = u + \frac{1}{4}\mathrm{Log}(1 + iz/B)$ has $q - u = O_M(1)$ and a fixed positive distance, depending on M, from the theta boundary. The exact fixed-point equation is

$$w = u + \frac{1}{4}\mathrm{Log}\left(1 + \frac{iz}{B} + \frac{6t(w^5-u^5) + \epsilon(w) - \epsilon(u)}{B}\right). \tag{19}$$

On the disk $|w-q| \leq C_M/u$, the logarithm argument stays in a right half-plane: its leading real part is at least one and its perturbation is $O_M(1/u)$. The derivative of the perturbation is $O_M(tu^4/B) = O_M(1/u)$. At q the displacement is $O_M(1/u)$; taking C_M large and then u large makes the disk invariant and the map contractive. The root is a full actual saddle with

$$\mathrm{Re}\, v = u + O_M(1), \qquad \mathrm{Re}\, e^{4v} \asymp_M B, \qquad e^{4\,\mathrm{Re}\, v} \asymp_M B.$$

In particular the initial theta margin is positive; the real stationarity equation strengthens it to $\mathrm{Re}\, e^{4v} \geq cta^5$ with an absolute $c > c_0$ for large u. Also $L/Q \leq C_M$.

For $y \leq -A_M/u$, the reflection bound (16) is at most $C_M e^{-A_M} + o(1)$, since $a - 1 \geq u/2$. Choose A_M large and then u large to make this less than $1/4$, while the relative error in (10) is less than $1/4$. Equation (15) excludes a zero. Conjugation therefore confines every zero in this region to $|\mathrm{Im}\, z| < A_M/u$.

For $x = \mathrm{Re}\, z$, $|x| \leq MB$, the same construction supplies an admissible real-frequency saddle. Once u is large enough that $A_M/u < 1/2$, Proposition 4.2 proves the zero real and simple. Every estimate is uniform in this bounded-ratio region. □

6 Actual far saddles and whole-plane coverage

Lemma 6.1. *There is an absolute C_0 such that, if $t \ge 1$, $X = |z|$, $\operatorname{Im} z \le 0$, $\alpha = \frac{1}{4}\log(X/(4\pi))$ is sufficiently large and*

$$X \ge C_0 t(\alpha+1)^5, \tag{20}$$

then z has an admissible actual saddle with real part $\alpha + O(1)$ and $e^{4a} \asymp X$.

Proof. Start with the leading model, not the actual kernel. The anchor $q = \frac{1}{4}\operatorname{Log}(iz/(4\pi))$ has real part α and imaginary part in $[-\pi/8, \pi/8]$, since $\operatorname{Re}(iz) \ge 0$. Solve

$$w = \tfrac{1}{4}\operatorname{Log}\{(iz + 6tw^5 + 9)/(4\pi)\}$$

on the disk $|w - q| \le Ct(\alpha+1)^5/X$. Choose C_0 so large that this radius is less than $1/100$. On the disk $|\operatorname{Im} w| < 1/2$, and $\operatorname{Re} w^5 \ge (\operatorname{Re} w)^5/2$ for large α. Hence the logarithm argument lies strictly in the right half-plane. Its modulus is comparable to X, and the map's derivative is $O(t(\alpha+1)^4/X) = O(1/\alpha)$. The center displacement is bounded by $Ct(\alpha+1)^5/X$, using $|\operatorname{Log}(1+h)| \le 2|h|$ for $|h| \le 1/2$. Increasing the fixed radius constant gives an invariant contraction disk. Its fixed point solves

$$6tw^5 + 9 - 4\pi e^{4w} = -iz.$$

If $w = a + ib$ and $Y = -\operatorname{Im} z$, its real part is

$$4\pi \operatorname{Re} e^{4w} = 6t(a^5 - 10a^3b^2 + 5ab^4) + 9 + Y \ge cta^5 + Y.$$

Since $|b| < 1/2$ and this quantity is positive, in fact $|b| < \pi/8$. Also $e^{4a} \asymp X$. Thus the model root lies strictly inside the theta strip with the required positive margin; boundary anchors cause no missing case.

Now restore the kernel error. The model derivative at its root is

$$30tw^4 - 16\pi e^{4w} = -4iz + t(30w^4 - 24w^5) - 36.$$

Its modulus is between cX and CX, by (20) with C_0 large. On a disk of radius $1/X$, the variation is $O(1)$, while ϵ and its first derivative are $O(e^{-4a})$. Subtracting the full slope divided by the central model derivative gives a contraction: its center displacement is $O(e^{-4a}/X)$, and its derivative is $O(1/X)$. The disk remains inside the strip because its boundary distance is at least $c(ta^5 + Y)/X \gg 1/X$. The resulting actual root preserves the positive margin; (7) restores the fixed admissibility constant. This proves the lemma. □

Proposition 6.2. *For an absolute sufficiently large C_1, the conditions $t \ge 1$, α sufficiently large and $|z| \ge C_1 t(\alpha+1)^5$ imply that every zero at z is real and simple. For each finite T, all nonreal zeros at $1 \le t \le T$ lie in one finite disk.*

Proof. Take $C_1 \ge 2C_0$. For $\operatorname{Im} z \le -1$, Lemma 6.1 and (16) exclude a zero. Conjugation excludes $\operatorname{Im} z \ge 1$. If $|\operatorname{Im} z| < 1$, let $x = \operatorname{Re} z$. For large $|z|$, $|x| \ge |z|/2$, and $\alpha_x = \frac{1}{4}\log(|x|/(4\pi)) \le \alpha$. Thus $|x| \ge C_0 t(\alpha_x + 1)^5$. Lemma 6.1 supplies a saddle at x. The local pair estimates in its proof hold on $|q - x| \le 1$. At any proposed zero there, the inequality $|\cos\Phi| \le \varepsilon \cosh \operatorname{Im}\Phi$ gives $|\operatorname{Im}\Phi| \le 2\varepsilon$. Vertical integration of $\Phi' = a + O(1/L)$ yields $|\operatorname{Im} z| \le 2\varepsilon/(a - O(1/L)) < 1/2$. Proposition 4.2 now applies. For $1 \le t \le T$, the displayed power condition holds uniformly for all sufficiently large $|z|$, since $(\log X)^5/X \to 0$. This proves the last assertion. □

Proof of Theorem 1.1. Use a real maximizing center u and $B = 4\pi e^{4u}$ from (18). If $X = |z| \geq MB$, write $\alpha = u + d$, so $d \geq \frac{1}{4}\log M$. Then

$$\frac{t(\alpha+1)^5}{X} \leq Ce^{-4d}\left(1 + \frac{d+1}{u}\right)^5 \leq C(2+d)^5 e^{-4d}, \qquad u \geq 1.$$

Choose one fixed $M > 1$ so large that the last expression is at most $1/C_1$ for every $d \geq \frac{1}{4}\log M$. For sufficiently late t, the center u and hence α are large, so Proposition 6.2 settles all $X \geq MB$. For this same fixed M, enlarge the time once more so Proposition 5.1 settles all $X \leq MB$. These regions cover the entire complex plane, using the same u and B. This proves the common-time conclusion. □

7 General integer-even orders: an expanding window

The whole-plane theorem above is sixth order. There is also a narrower result for every fixed positive integer m, which does not require the complex far-saddle argument. Set $n = 2m$ and define the actual transforms

$$F_{m,t}(z) = \int_0^\infty f_{m,t}(v)\cos(zv)\,dv, \qquad f_{m,t}(v) = e^{tv^n}K(v), \qquad M_{m,t} = F_{m,t}(0) > 0.$$

The same double-exponential kernel bound proves all required moments integrable, uniformly on compact parameter sets, for every fixed m. Thus these are entire, even, conjugation-compatible functions. Here the index $m = 1$ means a quadratic tilt and $m = 3$ the sixth-order tilt.

Theorem 7.1. *Fix $m \geq 1$. For all sufficiently large t, every maximizer u of $f_{m,t}$ on $[0,\infty)$ has the following property:*

$$|z| \leq u/2, \quad F_{m,t}(z) = 0 \quad \Longrightarrow \quad \operatorname{Im} z = 0, \quad F'_{m,t}(z) \neq 0.$$

Every such maximizing center tends to infinity with t. Consequently, for each fixed $R \geq 0$, all zeros in $|z| \leq R$ are real and simple at sufficiently late times. All constants and late times may depend on m; the last time may additionally depend on R.

We give complete actual-kernel estimates. The proof reorganizes the formal weighted-tail bounds with unspecified positive constants depending only on the fixed order; it does not replace the full kernel by its leading summand or assume a limiting probability law.

7.1 Real centers and an actual normalization bound

The positive continuous $f_{m,t}$ tends to zero at infinity, so attains a maximum $P = f_{m,t}(u)$. Every maximizer escapes: on $[0,R]$ its value is at most $C_R e^{tR^n}$ for $t \geq 0$, while at $R+1$ it is $K(R+1)e^{t(R+1)^n}$, eventually larger. This excludes every center in $[0,R]$ at once.

The real-axis specialization of (3), including its first two derivatives, gives for $v \to \infty$

$$L(v) := -\frac{K'(v)}{K(v)} = 4\pi e^{4v} - 9 + O(e^{-4v}), \qquad L'(v) = 16\pi e^{4v} + O(e^{-4v}).$$

Indeed write $K = 2\pi^2 e^{9v - \pi e^{4v}}(1+\rho)$ and differentiate $\log(1+\rho)$, whose first two derivatives are $O(e^{-4v})$. Choose one large $N \geq 2n$ such that $L > 0$ and

$$T(v) := \frac{L(v)}{nv^{n-1}} > 0, \qquad \frac{T'(v)}{T(v)} = \frac{L'(v)}{L(v)} - \frac{n-1}{v} \geq 3 \quad (v \geq N). \tag{21}$$

This is possible since the expression tends to four. At each sufficiently late maximum $u > N+1$, stationarity gives $t = T(u)$, whence

$$ce^{3u} \le t \le Qe^{4u} \tag{22}$$

for fixed $c, Q > 0$. The first bound integrates (21) from N to u; the second follows from $L(u) = O(e^{4u})$ and $nu^{n-1} \ge 1$. In particular $u \le Ct$ and $e^{2u} \le Ct$ at late centers. Moreover

$$(\log f_{m,t})'(v) = nv^{n-1}(t - T(v)), \qquad v \ge N.$$

Hence $f_{m,t}$ increases on $[N, u]$ and decreases to its right.

Normalization must use the actual mass $M = M_{m,t}$ rather than just the peak height. On $|v - u| \le 1$, the kernel derivative estimates and $t = T(u) = O(e^{4u}/u^{n-1})$ give

$$|(\log f_{m,t})''(v)| = |n(n-1)tv^{n-2} - L'(v)| \le Ce^{4u}.$$

Taylor's theorem at the stationary center implies $f_{m,t}(v) \ge Pe^{-C}$ for $|v - u| \le e^{-2u}$, after increasing C. This interval lies in the positive half-line. Integrating gives

$$M \ge c_1 e^{-2u} P, \qquad \frac{P}{M} \le Ce^{2u} \le Ct. \tag{23}$$

All these estimates hold uniformly over the actual maximizing centers.

7.2 Both tails with a growing exponential weight

For $s \ge 0$, (21) gives $T(u+s) \ge te^{3s}$. As $e^{3s} - 1 \ge 3s$, integration of the log slope yields

$$f_{m,t}(u+s) \le Pe^{-ts^2}. \tag{24}$$

Here the sharper exponent from integration is $-(3n/2)tu^{n-1}s^2 \le -ts^2$. For $0 < r \le 1/2$ and $0 \le s \le r$, one has $T(u-s) \le te^{-3s}$ and $1 - e^{-3s} \ge s$. For the latter inequality, $1 - e^{-3s} - s$ is concave, zero at zero and positive at $1/2$. Therefore

$$\log P - \log f_{m,t}(u-r) \ge nt(u-r)^{n-1} \int_0^r s\, ds \ge tr^2,$$

since $u - r \ge N \ge 2$ and $n \ge 2$. By monotonicity, the same upper envelope Pe^{-tr^2} bounds $f_{m,t}$ on $[N, u-r]$. On $[0, N]$, use $P \ge f_{m,t}(N+1)$ to obtain

$$\int_0^N f_{m,t}(v)\, dv \le CPe^{-t\Delta}, \qquad \Delta = (N+1)^n - N^n > 0.$$

Combining this with the interval length u and (23) gives the genuine left-mass bound

$$\frac{1}{M} \int_0^{u-r} f_{m,t}(v)\, dv \le C\{te^{-t\Delta} + t^2 e^{-tr^2}\}. \tag{25}$$

The second term uses $u \le Ct$; no compact prefix was discarded.

For $B \ge 0$ split the actual centered weighted first moment into $[0, u-r]$, $[u-r, u+r]$, and $u + (r, \infty)$. All these integrals are finite by the original double-exponential kernel envelope. On the left, $(u-v)e^{B(u-v)} \le ue^{Bu}$. On the middle interval, $|v-u|e^{B|v-u|} \le re^{Br}$ and its mass is at most M. On the right, (24) and $Bs \le B^2/(2t) + (t/2)s^2$ give

$$\int_r^\infty se^{Bs} f_{m,t}(u+s)\, ds \le \frac{P}{t} \exp\{B^2/(2t) - (t/2)r^2\},$$

because $\int_r^\infty s e^{-(t/2)s^2} ds = t^{-1} e^{-(t/2)r^2}$. Consequently

$$\frac{1}{M}\int_0^\infty |v-u| e^{B|v-u|} f_{m,t}(v)\, dv \le Cue^{Bu}\{te^{-t\Delta} + t^2 e^{-tr^2}\} + re^{Br} + C\exp\{B^2/(2t) - (t/2)r^2\}. \tag{26}$$

The constants do not depend on B or r within the stated ranges.

Choose $B = u$ and $r = e^{-u/2}$. Then $ur \to 0$ and $r^2 = e^{-u}$. The lower time bound gives $tr^2 \ge ce^{2u}$, whereas the upper bound controls $t^2 \le Q^2 e^{8u}$. For large u, $\Delta \ge e^{-u}$, so both exponentials in braces are at most $e^{-ce^{2u}}$ before their factors of t, t^2. The rightmost exponential tends to zero faster than e^{-u}, since $u^2/(2t) - (t/2)e^{-u} \le u^2/2 - (c/2)e^{2u}$. After multiplying (26) by u, the left and right tails still tend to zero; the middle term is $ue^{-u/2}e^{ue^{-u/2}} \to 0$. We have proved the crucial uniform actual moment conclusion

$$\frac{u}{M}\int_0^\infty |v-u| e^{u|v-u|} f_{m,t}(v)\, dv \longrightarrow 0. \tag{27}$$

It holds as $t \to \infty$ for every maximizing center, because every such center escapes and all preceding bounds are uniform at those centers.

7.3 Cosine approximation and exact real simple zeros

The elementary exponential estimate $|e^w - 1| \le |w| e^{|w|}$ implies, by applying it to the two terms of the cosine,

$$|\cos(zv) - \cos(zu)| \le e^{u|\operatorname{Im} z|}|z|\,|v-u| e^{|z||v-u|}.$$

Integrating and using (27) gives, for every $\eta > 0$, at one sufficiently late time independent of z,

$$\left|\frac{F_{m,t}(z)}{M} - \cos(uz)\right| \le \eta e^{u|\operatorname{Im} z|}, \qquad |z| \le u. \tag{28}$$

This is a growing-frequency estimate, not just compact convergence.

Take $\eta = 10^{-6}$ and $u \ge 2$. If z is a zero in $|z| \le u/2$, write $q = uz = a + ib$. The identity $|\cos(a+ib)|^2 = \cos^2 a + \sinh^2 b$ and (28) give $\sinh|b| \le \eta e^{|b|}$. Thus $1 - e^{-2|b|} \le 2\eta$, whence $|b| \le 2\eta$. Define $g(w) = F_{m,t}(w/u)/M$. On $|w-a| \le 1$,

$$|w/u| \le u/2 + 1/u \le u, \qquad |g(w) - \cos w| \le e\eta.$$

Indeed $|a| \le |q| \le u^2/2$ and $|\operatorname{Im} w| \le 1$. Cauchy's derivative bound on circles of radius $1/2$ gives $|g'(w) + \sin w| \le 2e\eta$ for $|w-a| \le 1/4$. Since $g(q) = 0$, the cosine bound at q and the derivative bound $|\sin w| \le e$ on the unit-height strip imply $|\cos a| \le \eta e^{2\eta} + 2e\eta < 10\eta$. Hence $|\sin a| > 1/2$. On the disk $|w-a| \le 10\eta$,

$$|g'(w) + \sin a| \le 2e\eta + 10e\eta < 1/4 < |\sin a|/2.$$

Integrate g' along any line segment in this disk. Subtracting the constant derivative $-\sin a$ gives

$$|g(w_1) - g(w_2)| \ge \tfrac{1}{2}|\sin a|\,|w_1 - w_2|.$$

Both q and $\bar{q}$ lie in the disk and are zeros by conjugation compatibility. The lower bound forces $q = \bar{q}$. It also gives $g'(q) \ne 0$, and $g'(q) = F'_{m,t}(z)/(uM)$ proves simplicity. This proves the expanding-window assertion. For a fixed R, take a late time such that every center has $u \ge 2R$; the fixed-disk assertion follows with the same zero argument.

7.4 The exact quantifier boundary

For each fixed m, R there is a late time valid on that disk. The argument does not give a time uniform over every R, nor establish global reality for every integer-even order. Any nonreal zeros, if they exist at late times, must leave each fixed disk. This is not an existence assertion about such zeros. The elementary polynomials $p_t(z) = z^2 + t^2$, $t > 0$, have no zeros in $|z| \le t/2$, yet have the nonreal roots $\pm it$. They are a counterexample to inferring global reality from expanding zero-free windows alone, not models of the arithmetic transform. Their root formula and $|it| = t$ prove both statements immediately. The sixth-order whole-plane theorem requires the separate far-frequency work above. No $A(2)$/RH implication or interpolation of reality thresholds is supplied.

8 Exact formal correspondence and publication scope

The frozen mathematical baseline is

`bab688bb6140cd6316c80680095a914ea0d346db.`

The exact final declaration is

`RiemannSixthTransitionSimpleZeros.`
`eventually_all_zeros_real_simple.`

Its namespace prefix is

`OperatorDerivativeRewrite.Foundation.`

The formal transform is

`RiemannEvenFlow.flow3tz`:

the index three means a sixth power, not a third power. The proof combines the bounded-ratio theorem with

`RiemannSixthFarZeroControl.`
`uniform_actual_far_zeros_real_simple`,

both at a single stationary center. The theorem map records exact source lines and hashes. The human proof consolidates explicit formal constants into uniform inequalities; intermediate regrouped lemmas are not claimed as new named formal statements.

De Bruijn [2] and Newman [3] provide the primary historical comparison for real-zero Fourier transforms. Newman's quadratic propagation theorem does not supply the sixth-order conclusion. Targeted priority review remains necessary; novelty is unresolved.

The theorem gives an existential late time. It gives no numerical optimal threshold, no nonreal-zero certificate at a specified early sixth-order time, no assertion that the set of all reality times is an upper ray, and no threshold collision. The whole-plane argument is presented for the verified sixth-order transform only. Section 7 proves the narrower general integer-even expanding-window result. The source chain runs through `RiemannEvenGrowingMoment`, `RiemannEvenGrowingFourier` and `RiemannEvenWindowZeros`; its exact declarations and the classical human proof reorganizations are in the map. Constants and times are order dependent. No global integer-order classification, real-order interpolation or Riemann-hypothesis implication follows.

References

[1] Project Stardust, *An attained eventual reality threshold for a growing quartic arithmetic transform*, reviewed editorial draft, October 29, 2026. Companion manuscript in `manuscripts/quartic-threshold/`.

[2] N. G. de Bruijn, The roots of trigonometric integrals, *Duke Math. J.* **17** (1950), 197–226. doi:10.1215/S0012-7094-50-01720-0.

[3] C. M. Newman, Fourier transforms with only real zeros, *Proc. Amer. Math. Soc.* **61** (1976), 245–251. `https://sites.math.northwestern.edu/~auffing/papers/Newman.pdf`.

[4] NIST Digital Library of Mathematical Functions, equation 20.7.32, modular theta identity. `https://dlmf.nist.gov/20.7.E32`.